Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 11-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-16** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**A**	2	**B**	3	**C**	4	**C**	5	**BC**	6	**BD**
7	**ABC**	8	**ABC**	9	**7**	10	**5**	11	**8**	12	**6**
13	**9**	14	**4**	15	**2**	16	**5**				

PHYSICS

17	**A**	18	**C**	19	**D**	20	**C**	21	**ACD**	22	**ABC**
23	**AC**	24	**BC**	25	**7**	26	**8**	27	**7.2**	28	**9**
29	**5**	30	**5**	31	**4**	32	**8**				

CHEMISTRY

33	**D**	34	**C**	35	**C**	36	**C**	37	**CD**	38	**ABCD**
39	**ABC**	40	**AC**	41	**9**	42	**78**	43	**2**	44	**6**
45	**7**	46	**10**	47	**8**	48	**4**				

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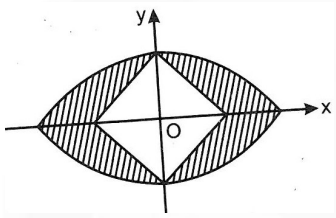
SOLUTIONS

MATHEMATICS

1. $f(x)=\lim\limits_{n\to\infty}\tan^{-1}\left(4n^2.2\sin^2\frac{x}{2n}\right)=\lim\limits_{n\to\infty}\tan^{-1}\left(8n^2\left(\frac{\sin\frac{x}{2n}}{\frac{x}{2n}}\right)^2.\frac{x^2}{4n^2}\right)=\tan^{-1}\left(2x^2\right)$

$$g(x)=\lim_{n\to\infty}\frac{n^2}{2}\left(\frac{\ln\left(1+\cos^2\frac{2x}{n}-1\right)}{\cos^2\frac{2x}{n}-1}\right)\left(\cos\frac{2x}{n}-1\right)=x^2$$

2. $|x|+|y|\geq 2$ and $\frac{x^2}{9}+\frac{y^2}{4}=1$

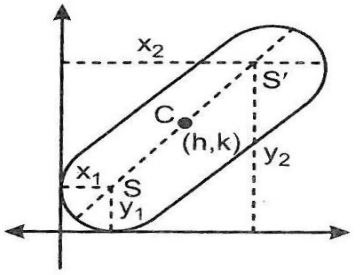


Ar. of ellipse – Ar. Of square $=\pi(2)(3)-8=6\pi-8$

3. The director circle of the parabola is its directrix (x + 11 = 0). Now apply condition of tangency.

4. $f(x)=2\sin^2 x+\sin x+2\forall x\in\left[\frac{\pi}{6},\frac{2\pi}{3}\right]$ Let $\sin x=t$ $f(t)=-2t^2+t+2\forall t\in\left[\frac{1}{2},1\right]$

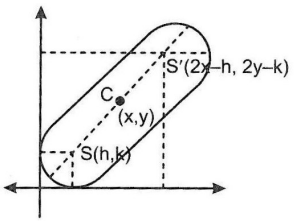
5. $SS'=2ae$ $\left(x_1-x_2\right)^2+\left(y_1-y_2\right)^2=4(2)^2\left(\frac{\sqrt{3}}{2}\right)^2$



$\left(x_1+x_2\right)^2+\left(y_1+y_2\right)^2-4\left(x_1x_2+y_1y_2\right)=12$

($\because x_1x_2$ and y_1y_2 are $\perp$ distinct of the foci from their tangents = $b^2=1^2$)

$\Rightarrow h^2+k^2=5$ $(2x-h)(h)=1\Rightarrow x=\frac{1+h^2}{2h}$



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$(2y-k)(k)=1 \Rightarrow y=\dfrac{1+k^2}{2k}$

6. $a=(cp+r)^2$ and $b=(cp-r)^2$

7. $f'(x)=\dfrac{\cos x}{x}$ clearly, $f'(x)>0 \Rightarrow x \in \left(0, \dfrac{\pi}{2}\right) \cup \left((4n-1)\dfrac{\pi}{2}, (4n+1)\dfrac{\pi}{2}\right) \forall n \in N$

and $f'(x)<0 \Rightarrow x \in \left((4n+1)\dfrac{\pi}{2}, (4n+3)\dfrac{\pi}{2}\right) \forall n \in N$ $\therefore f(x)$ has a local minima at x

$= (4n-1)\dfrac{\pi}{2} \forall n \in N$ And f(x) has a local maxima at $x=\dfrac{\pi}{2}$ and $(4n+1)\dfrac{\pi}{2} \forall n \in N$

Also, $f''(x)=\dfrac{x(-\sin x)-\cos x}{x^2}=0 \Rightarrow x \tan x+1=0$

$\therefore$ all the points of inflection of f(x) lie on the curve x tanx + 1 = 0

also, $f'(x)=0 \Rightarrow x=(2n+1)\dfrac{\pi}{2} \forall n \in N$

$\therefore$ number dof values of 'x' in $(0, 10\pi)$ in which $f'(x)=0$ are 20

8. $4\sin 3x+5 \geq 4\cos 2x+5\sin x$ $\Rightarrow (\sin x-1)(4\sin x+1)^2 \leq 0 \forall x \in R$

9. $D=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 2 & 5 & \lambda\end{vmatrix}=\lambda-8$ $D_3=\begin{vmatrix}1 & 1 & 2\\ 1 & 2 & 14\\ 2 & 5 & \mu\end{vmatrix}=0$

10. $a_n=a_1+(n-1)d+\dfrac{1}{2}(n-1)(n-2)$

So a_n is a quadratic polynomial in 'n' So $a_n=\dfrac{(n-\alpha)(n-\beta)}{2}$

since $a_{19}=a_{92}=0$ we must have $a_n=\dfrac{1}{2}(n-19)(n-92)$

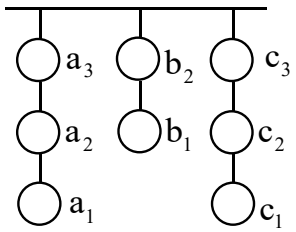
so $a_3=\dfrac{1}{2}(3-19)(3-92)=712$

11. f(2 – x) = f(2 + x), it means it symmetric about x = 2 $\Rightarrow \int_0^2 f(x)\,dx=\int_2^4 f(x)\,dx=5$

Let $2-x=t; f(t)=f(4-t)$ i.e., $f(x)=f(4-x)=f(4+x)$

$\int_0^{50} f(x)\,dx=25\left(\int_0^2 f(x)\,dx\right)=25\times 5=125$

12.



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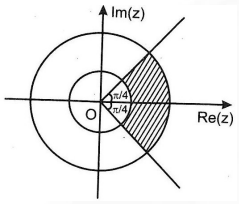
$(a_1, a_2, a_3), (b_1, b_2)$ and (c_1, c_2, c_3) are alike things so these can be arranged is

$$\frac{8!}{2!3!3!} = \frac{4.5.6.7.8}{2\times 6} = 560$$

13. $f(n) = {}^nC_0\, a^{n-1} - {}^nC_1\, a^{n-2} + \ldots\ldots \quad \Rightarrow f(n) = \frac{(a-1)^n}{a}$

$$f(2007) + f(2008) = 3^7 K \quad \Rightarrow \frac{3^9 + (a-1)3^9}{a} = 3^7 K \Rightarrow K = 9$$

14.



$2 \le |z| \le 4 \qquad$ Probability $= \frac{1}{4}$

15. $a^2 \sec^2 200^0 = c^2 \tan^2 200^0 + d^2 + 2cd \tan 200^0$

$b^2 \sec^2 200^0 = c^2 + d^2 \tan^2 200^0 - 2cd \tan 200^0 \quad \Rightarrow a^2 + b^2 = c^2 + d^2$

$(a\sec 200^0 - c\tan 200^0)^2 = d^2 \qquad (b\sec 200^0 + d\tan 200^0)^2 = c^2$

$\Rightarrow (c^2 + d^2)(\sec^2 200^0 + \tan^2 200^0) + (2bd - 2ac)\sec 200^0 \tan 200^0 = c^2 + d^2$

$$\Rightarrow \frac{2(c^2+d^2)}{ac-bd} = \frac{2\sec 200^0}{\tan 200^0} = \frac{2}{\sin 200^0} = \frac{-2}{\sin 20^0}$$

16. $g(f(x)) = x$

$g'(f(x))f'(x) = 1$

$f(1) = -\frac{7}{6} \qquad \therefore x = 1$

$g'\left(-\frac{7}{6}\right)f'(1) = 1 \qquad g'\left(-\frac{7}{6}\right) = \frac{1}{f'(1)}$

$h(x) = ax^{-\frac{5}{4}} + bx^{\frac{1}{4}} \qquad h'(x) = \frac{5a}{4}x^{\frac{-9}{4}} + \frac{b}{4}x^{\frac{-3}{4}}$

$h'(5) = 0 \Rightarrow \frac{5a}{4}.5^{\frac{-9}{4}} + \frac{b}{4}.5^{\frac{-3}{4}} = 0$

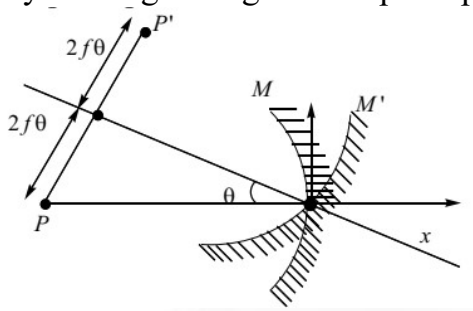
$\Rightarrow 5a.5^{-3/2} = b \qquad \Rightarrow \frac{a}{b} = 5^{1/2}$

$\left(\frac{a}{b}\right)^2 = 5 \qquad \frac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} = \frac{5}{5\times\frac{1}{5}} = 5$

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PHYSICS

17. By rotating an angle θ the principal axis of mirror also rotated by same angle.



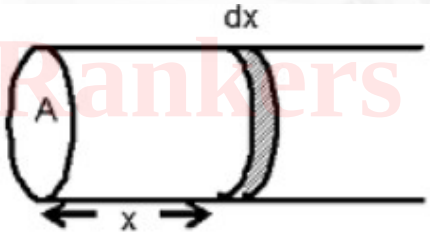
18. $V=\pi a^2 L \;\; dV=(2\pi aL)da \qquad\qquad \Delta V=2\pi aL\,(\Delta a)$

$$B=\frac{-\Delta p}{\left(\frac{dV}{V}\right)},\ \Delta p=\frac{2B\Delta a}{a}\Rightarrow F=\mu(\Delta)\times \text{surface area}$$

$$F=\mu\frac{2B\Delta a}{a}(2\pi aL)=4\pi\mu B\,L\Delta a$$

19. f_1 by $O_1=f_0\left(\frac{V}{V-\frac{V}{5}}\right)=f_0\frac{5}{4}\lambda$ received by $O_1=\frac{v}{f_1}=\frac{4v}{5f}f_2$ by $O_2=f_0\left(\frac{5}{4}\right)\left(\frac{4v+\frac{v}{5}}{4v}\right)=\frac{21f}{16}$

20.



(i) $dR=\frac{1}{\sigma(x)}\frac{dx}{A}=\frac{\sqrt{x}dx}{Al\sigma_0} \qquad \therefore R=\int_0^l dR=\frac{1}{Al\sigma_0}\int_0^l\sqrt{x}\,dx=\frac{2\sqrt{l}}{3A\sigma_0}$

(ii) $\frac{V_0}{R}=\frac{3A\sigma_0 V_0}{2\sqrt{l}}$ The current density, $J=\frac{1}{A}=\frac{3\sigma_0 v_0}{2\sqrt{l}}$

Electric field, $E(x)=\frac{J}{\sigma(x)}=\frac{3V_0\sqrt{x}}{2l^{3/2}}$

21. The radius of curvature : $R=\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left(\frac{d^2y}{dx^2}\right)}$, radius of curvature at $(0,-b)$ is $\frac{a^2}{b}$, radius of curvature at $(-a,0)$ is $\frac{b^2}{a}$, acceleration of the particle $(0,-b)$ is $\frac{V_0^2 b}{a^2}$ angular momentum about the origin when the particle (0, b) is $mV_0 b$

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22. Energy density $= \frac{1}{2}\in_0 E^2$, $E_{infinite} = \frac{kQx}{R^3}$, $E_{outside} = \frac{kQ}{r^2}(r > R)$

23. Resolve the electric dipole into components along the diagonal of the square

$F_x = \frac{6kpq}{r^3}\cos\theta \quad F_y = \frac{6kpq}{r^3}\sin\theta \quad F_R = \frac{6kpq}{r^3}$

24. Conceptual

25. The resistance shorting the capacitor plates may be considered in parallel with the capacitor and at time t = 0 the capacitor behaves as a short circuit

26. Radius of curvature of the curve $r = \frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx}}$

$y = 2x^2, \frac{dy}{dx} = 4x, \frac{d^2y}{dx^2} = 4 \qquad r = \frac{1+16x^2}{4} \Rightarrow \text{at } x = 0, r = \frac{1}{4}$

$r = \frac{mv}{qB} \Rightarrow B = \frac{mV}{qr} = 8$

27. Conceptual

28. Pitch = 0.1 mm, N = 100

Least count $= \frac{0.1}{100} = 0.001$ mm

Negative zero error $= 6 \times 0.001 = 0.006$ mm

Reading of screw gauge $= (50 \times 0.1)\text{ mm} + (30 \times 0.001)\text{ mm} = 5.03$ mm

Measures values = (5.03 + 0.006)mm = 5.036 mm

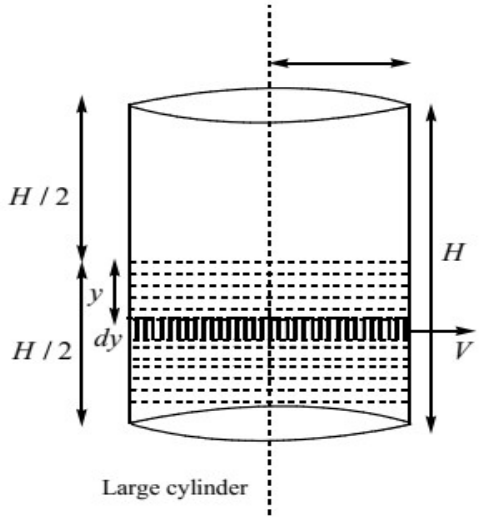
29. $v = \sqrt{2gy}$.....(i)

Consider a strip at depth 'y'

Volume flow rate from strip

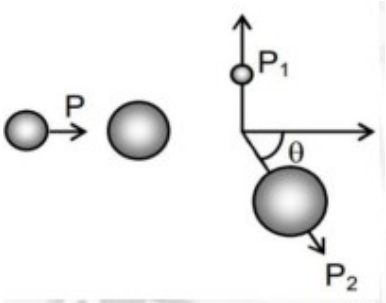
$dQ = av = k(2\pi R)dy\sqrt{2gy}$

Net volume flow are $Q = k2\pi R\sqrt{2g}\int_0^{H/2}\sqrt{y}\,dy \; Q = \frac{\pi}{500}R\sqrt{g}H^{3/2}$



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30.



$P_2\cos\theta = P$ .....(i)

$P_2\sin\theta = P_1$ .....(ii)

Solving (i) and (ii), we get

$P_2^2 = P^2 + P_1^2$ $2m_2k_2 = 2mk + 2m_1k_1$

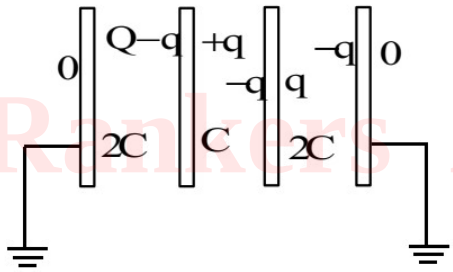
$10k_2 = 4k + k_1$ $10k_2 - k_1 = 4k$

$10k_2 - k_1 = 56$.....(iii)

$k_1 + k_2 = 14 - 2.9$ $k_1 + k_2 = 11.1$ .....(iv)

Solving (iii) and (iv), we get $\Rightarrow k_2 = 67.1 \Rightarrow k_2 = 6.1$ MeV $k_1 = 5.0$ MeV

31. $\frac{q}{2C} + \frac{q}{C} + \frac{q-Q}{2C} = 0 \Rightarrow 2q = \frac{Q}{2} \Rightarrow q = +\frac{Q}{4}$



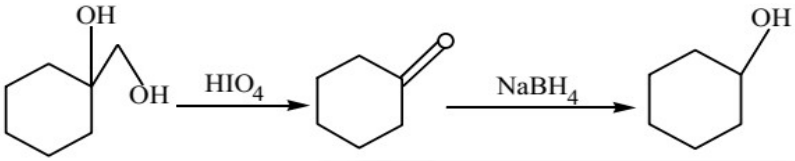
32. $k_1 = \frac{5}{4}\text{N/cm},\ k_2 = \frac{5}{1}\text{ N/cm},\ F = (k_1 + k_2)x \Rightarrow x = \frac{5N}{\left(\frac{5}{4} + 5\right)\text{N/cm}} = 0.8\text{ cm}$

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CHEMISTRY

33. Intra Hoffmann bromide degradation

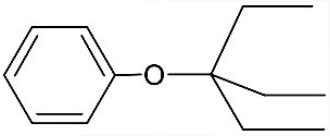
34.



35. Lesser value of $E^0_{Ni^{+2}/Ni}$ is due to high hydration energy.

36. For zero order reaction change in concentration is same in equal interval of time

37.



undergoes S_N1 reaction

38. I) $2PbO + 4HCl \rightarrow 2PbCl_2 + 2H_2O$ and $PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + 2H_2O$

II) $Mg(g) \rightarrow Mg^{2+}_{(g)}, \Delta H = +ve$

$O(g) \rightarrow O^{2-}_{(g)}, \Delta H = +ve$

$Mg^{2+}_{(g)} + O^{2-}_{(g)} \rightarrow MgO(s), \Delta H = -ve$

39. I) $\left[Pt\,Cl_2(en)_2\right]^{2+}$ has three stereo isomers

II) $\left[NiCl_4\right]^{2-}$ only paramagnetic

III) $\left[Fe(H_2O)_6\right]^{2+}$: $3d^6$ highest s pin only magnetic moment

40. Conceptual

41. Actual pressure $\rightarrow 725 - 25 = 700$ mm

Volume of nitrogen $= \dfrac{273 \times 700 \times 44}{300 \times 700} = 33.52$ ml

22400 ml of nitrogen at STP weight = 289

So 33.52 ml of N_2 ast STP weight $= \dfrac{28 \times 33.52}{22400} = 0.0419$ gm

% of N_2 in organic compound $= \dfrac{0.0419}{0.25} \times 100 = 18.4$

42. (Given atomic masses C = 12, K = 1.0 , O = 16.0, Br = 80.04)

Moles of benzoic acid $= \dfrac{6.1}{122}$ = moles of m-bromo benzoic acid.

So, weight of m-bromo benzoic acid $= \dfrac{6.1}{122} \times 201g = 10.05g$

$$\% \text{ yield} = \frac{\text{actual weigth}}{\text{theretical weight}} \times 100$$

$$= \frac{7.8}{10.05} \times 100 = 77.61\% = 78\%$$

43. Conceptual

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44. I) $Ag_2O \xrightarrow{\Delta} Ag + O_2\uparrow$ II) $KClO_3 \xrightarrow[\Delta]{MnO_2} KCl + O_2\uparrow$

III) $Pb_3O_4 \xrightarrow{\Delta} PbO + O_2\uparrow$

IV) $Pt + HNO_3 + HCl \rightarrow PtCl_6^{-2} + NO + H_2O$

V) $KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

VI) $NaNO_{2(s)} + \text{dil. } H_2SO_4 \longrightarrow Na_2SO_4 + NO_2\uparrow$

45. $K_{sp}\left[Zn(OH)_2\right] = \left[Zn^{+2}\right]\left[OH^-\right]^2 \quad 2\times10^{-13} = 0.2\times\left[OH^-\right]^2$

$\left[OH^-\right] = 10^{-6} \quad K_{sp}\left[Al(OH)_3\right] = \left[Al^{+3}\right]\left[OH^-\right]^3$

$16\times10^{-25} = \left[Al^{+3}\right]\left[10^{-6}\right]^3 \quad \left[Al^{+3}\right] = 16\times10^{-7}$

46. $Mg \rightarrow Mg^{2+} + 2e^-$

$$\frac{Cu^{2+} + 2e^- \rightarrow Cu}{Mg + Cu^{2+} Mg \rightarrow Mg^{2+} + Cu}$$

$$E = 2.67 = 2.7 - \frac{RT}{nF}\ln\frac{x}{1}$$

$$0.03 = \frac{300}{2\times11500}\ln x$$

$2.3 = \ln x$

$X = 10$

47. Mass of oxide = 2 gm

Mass of metal = 1.6 gm

Mass of oxygen = 0.4 gm

$$\text{Eq. wt of metal} = \frac{16\times8}{0.4} = 32$$

48. $X_B = 0.5 \; X_T = 0.5$

$P_{total} = P_B^0 X_B + P_T^0 . X_T$

$80 = 0.5P_B^0 + 0.5P_T^0$

$160 = P_B^0 + P_T^0 .....(1)$

$$y_B = \frac{P_B^0 . X_B}{P_{total}} = \frac{P_B^0 \times 0.5}{80} = \frac{P_B^0}{160}$$

For condensate

$$P_{total} = 100 = P_B^0 . y_B + P_T^0 . y_T = 100 = P_B^0 . \frac{P_B^0}{160} + P_T^0 . \frac{P_T^0}{160}$$

$P_B^{0^2} + P_T^{0^2} = 100\times160.....(2)$

From (1) and (2)

$P_B^0 = 120 \qquad P_T^0 = 40$

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